# Netbooting a Large Language Model-based OS in an Ubuntu Live Server


This brief tutorial explores the wireless netbooting of the LLM model Gemini AI in an Ubuntu server.

With the rapid advancement of artificial intelligence, large language models (LLMs) require specialised operating systems that optimise resource utilisation, ensure efficient model execution, and facilitate seamless deployment across various hardware architectures. An LLM OS is designed to meet these demands by providing a streamlined environment tailored for AI workloads, incorporating high-performance computing, distributed processing, and low-latency networking. Given the extensive computational and storage requirements of LLMs, netbooting presents a practical solution for deploying an LLM OS without relying on local storage. By leveraging network-based booting mechanisms such as PXE (Preboot Execution Environment), TFTP (Trivial File Transfer Protocol), and NFS (Network File System), an LLM OS can be centrally managed, rapidly provisioned, and dynamically updated. This approach not only reduces hardware dependencies but also ensures uniformity across multiple AI nodes, making it ideal for large-scale AI research, data centres, and enterprise deployments.

## AI and LLM integration with netbooting

AI-based LLMs are enabling advanced natural language processing (NLP) capabilities that help organisations extract insights from massive datasets, automate content generation, and enhance customer support systems. Deploying an OS that directly boots into an interactive TTY-based LLM model can eliminate unnecessary overhead from traditional installations and significantly reduce boot times. By leveraging containerised environments such as Docker or Singularity, organisations can dynamically load AI models like Llama 2, GPT-based models, or BERT directly from network-attached storage without requiring individual local installations.

Furthermore, using a stateless boot model for LLM processing ensures that each session runs in a clean environment, preventing data persistence issues and improving security by mitigating malware risks. This approach is particularly useful in GPU-based AI clusters, where machines need to be reinitialised frequently for different workloads.